

2.4 Nitrogen & Sulfur

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.4 Nitrogen & Sulfur
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What happens when silver chloride dissolves in aqueous ammonia?

- 1 Ammonia forms a complex with the Ag^+ ion.
- 2 A redox reaction occurs.
- 3 Ammonia acts as a Brønsted-Lowry base.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 2

Oxygen and nitrogen are consecutive elements in Period 2 and both form diatomic molecules.

The bond length in the O_2 bond is 1.21 Å and the bond length in N_2 is 1.09 Å.

What best explains the difference in bond length?

- A. Oxygen atoms are more electronegative.
- B. Nitrogen has more bonding electrons.
- C. Nitrogen has a smaller atomic radius.
- D. Oxygen bond is stronger.

[1 mark]

Question 3

Which statement about ammonia is completely correct?

- A. Ammonia acts as a nucleophile by accepting a pair of electrons when it reacts with bromoethane.
- B. Ammonia can form a co-ordinate bond with a hydrogen ion to form an ammonium ion.
- C. Ammonia is a base and accepts hydroxide ions.
- D. The shape of the ammonia molecule is pyramidal with bond angles of 109.5°

[1 mark]

Question 4

“Photochemical smog is produced when 1 acts on 2 and 3.”

Which row correctly identifies 1, 2 and 3?

	1	2	3
A	Heat	Nitrogen	Oxygen
B	Sunlight	Nitrogen oxides	Sulfur oxides
C	Sunlight	Nitrogen oxides	Hydrocarbons
D	Heat	Hydrocarbons	Oxygen

[1 mark]

Question 5

Lime, CaO , is used to reduce the acidity of soil, and ammonium nitrate is a nitrogenous fertiliser.

Why can they not be used in a mixed form?

- A. $\text{Ca}(\text{NO}_3)_2$ is produced, which hardens water.
- B. The dry mixture is explosive.
- C. If damp, ammonia is given off.
- D. If damp, nitric acid will form.

[1 mark]

Question 6

In which of the following reactions does the product produced turn damp litmus paper blue?

- 1 $\text{NH}_4\text{NO}_3(\text{s}) + \text{O}_2(\text{g})$
- 2 $\text{NH}_4\text{Cl}(\text{s}) + \text{NaOH}(\text{aq})$
- 3 $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 7

Atmospheric oxides and unburned hydrocarbons cause the production of peroxyacetyl nitrate (PAN).

Which of the following is **unaffected** by this reaction?

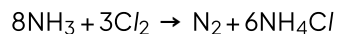
- 1 Corrosion of architecture.
- 2 Lung irritation and difficulty breathing.
- 3 Reduction of air clarity.

- A. 1 only
- B. 1 and 2
- C. 2 only
- D. 2 and 3

[1 mark]

Question 8

Chlorine react with ammonia in the gas phase.



Which statements are correct?

- 1 Ammonia behaves as a reducing agent.
- 2 The oxidation number of the hydrogen changes.
- 3 Ammonia behaves as a base.

A. 1 only

B. 1 and 2

C. 1 and 3

D. 1, 2 and 3

[1 mark]

Question 9

Farmers often use ammonium chloride as a fertiliser to supply the growing plants with nitrogen in the form of ammonia.

Which of the following would reduce ammonium chloride's effectiveness as a fertilizer?

- 1 Calcium carbonate, CaCO_3
- 2 Magnesium hydroxide, $\text{Mg}(\text{OH})_2$
- 3 Sodium sulfate, Na_2SO_4

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 10

Modern cars have catalytic converters fitted to their exhaust pipes. Despite this, analysis of exhaust gases shows small quantities of nitrogen oxides.

Which amendments would result in lower concentrations of nitrogen oxides in exhaust fumes?

- 1 Increasing the surface area of the catalyst in the converter.
- 2 Increasing the rate of flow of the exhaust gases through the converter.
- 3 A much higher temperature of combustion in the engine.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Model Answers: Medium

1

The correct answer is **D** because:

- Aqueous ammonia is used as a secondary test for the **identity of halides** after they have reacted with silver nitrate to produce a silver halide precipitate.
- In the reaction, ammonia **forms a complex** with the **Ag⁺** ion:
 - $\text{Ag}^+\text{Cl}^- + 2\text{NH}_3 \rightarrow [\text{Ag}(\text{NH}_3)_2]^+ + \text{Cl}^-$
- Therefore, **AgCl/dissolves** which shows **chloride ions** are present instead of bromide (only dissolves in concentrated ammonia) or iodide (doesn't dissolve).

A, B & C are incorrect as the oxidation number of all atoms in the reaction remains the same, therefore it is not redox reaction there is no movement of protons between species. So nothing acts as a Bronsted-Lowry acid or base.

2

The correct answer is **B** because:

- There is a **double bond** (4 bonding electrons) between oxygen atoms in O₂ and a **triple bond** (6 bonding electrons) between nitrogen atoms in N₂.
- The nitrogen triple bond has more **bond strength** and therefore a stronger the pull between the two atoms .
- Stronger bonds are shorter bonds.

A is incorrect as electronegativity causes a greater pull between the nucleus and the bonding electrons and therefore decreases bond length.

C is incorrect as nitrogen has a larger atomic radius than oxygen because nitrogen has less protons so the pull between the outer electrons and the nucleus is less.

D is incorrect as the nitrogen bond is stronger because it is a triple bond.

3

The correct answer is **B** because:

- Ammonia (NH_3) contains a **lone pair** of electrons on the nitrogen atom.
- This lone pair of electrons can be donated to a hydrogen ion (H^+) which doesn't have any electrons to form a **co-ordinate bond**.
- Co-ordinate (also called dative) bonds are made of **two electrons from the same atom** being shared by both atoms in the bond, rather than a normal covalent bond which is formed from one electron from each atom being shared.
- When accepting a hydrogen ion (also called a proton) ammonia acts as a **Bronsted-Lowry base** – a **proton acceptor**.

A is incorrect as nucleophiles **donate** a pair of electrons.

C is incorrect as bases accept **hydrogen** ions.

D is incorrect as ammonia has a trigonal pyramidal shape and **bond angles of 107°** , this is less than 109.5° because a lone pair of electrons exert **more repulsion** than bonding pairs.

4

The correct answer is **C** because:

- **Nitrogen** and **unburnt hydrocarbons** are pollutants from burning **fossil fuels**.
- **Sunlight** (UV light) causes atmospheric oxides of nitrogen (NO_2 and NO) react with unburnt hydrocarbons to form **peroxyacetyl nitrate** (PAN).
- PAN is a component of **photochemical smog**.
- Photochemical smog causes haze in the atmosphere, visual and respiratory problems.

You are not expected to know the reaction pathway or equation, only the facts above.

5

The correct answer is **C** because:

- Fertilisers are used to provide plants with nitrogen, required for growth, in the form of ammonium salts.
- If calcium oxide is added to water it forms calcium hydroxide, which is a **strong base**.
- When an ammonium compound reacts with a base ammonia is released:
 - $\text{NH}_4\text{NO}_3 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{NH}_3 + 2\text{H}_2\text{O} + \text{Ca}(\text{NO}_3)_2$
- Therefore if the two are added together in a solution **ammonia gas** is evolved, **wasting the fertiliser** (the useful part is lost to the atmosphere).

A is incorrect as calcium nitrate is formed but the hardening of water does not affect the fertilisation of plants.

B is incorrect as ammonium nitrate is very explosive but this does not explain why it can't be added to soil with calcium oxide.

D is incorrect as calcium nitrate is formed not nitric acid.

6

The correct answer is **C** because:

- Litmus paper turns **blue** in contact with a **base** (a proton acceptor).
- Reaction 2: $\text{NH}_4\text{Cl}(\text{s}) + \text{NaOH}(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) + \text{NaCl}(\text{aq})$
 - **NH₃ is basic** (will accept protons) so therefore turns damp litmus paper blue.
- Reaction 3: $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NH}_4\text{OH}(\text{aq})$

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- Reaction 3: $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NH}_4\text{OH}(\text{aq})$
 - NH_4OH is basic / alkaline because it dissociates into its ions to produce OH^- .
 - **OH⁻ is basic** (will accept protons) so therefore turns damp litmus paper blue.

A, B & D are incorrect as $\text{NH}_4\text{NO}_3(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$. None of the products are basic or alkaline.

7

The correct answer is **A** because:

- Peroxyacetyl nitrate (PAN) is a constituent of **photochemical smog**.
- Photochemical smog causes **lung irritation** and **reduction of air clarity (haze)**.

Corrosion of architecture is caused by acid rain, this is when atmospheric oxides (SO_2 and NO_2) react to form SO_3 which dissolved in water to form sulfuric acid, H_2SO_4 . Sulfuric acid corrodes architecture, acidifies lakes and kills plants.

8

The correct answer is **C** because:

- **Reducing agents** cause another species to be reduced (have a lower oxidation number) whilst being **oxidised itself** (increasing in oxidation number).
- In this reaction the oxidation numbers are as follows:

8NH ₃		+	3Cl ₂		→	N ₂		+	6NH ₄ Cl		
N	H		Cl			N			N	H	Cl
-3	+1		0			0			-3	+1	-1

- **Nitrogen increased** in oxidation number from -3 in NH₃ to 0 in N₂.
- **Chlorine decreased** in oxidation number from 0 in Cl₂ to -1 in NH₄Cl.
- Ammonia accepts a hydrogen ion (also called a proton) to form an ammonium ion (NH₄⁺) and therefore act as a **Bronsted-Lowry base** – a **proton acceptor**.

B & D are incorrect as the oxidation number of hydrogen does **not** change in this reaction, it remains +1.

9

The correct answer is **B** because:

- Fertilizers are used to provide plants with nitrogen, required for growth, in the form of ammonium salts.
- If an ammonium compound is reacted with a base **ammonia gas** is evolved, **wasting the fertilizer** (the useful part is lost to the atmosphere).
- **ammonium compound + base → salt + ammonia + water** (+ carbon dioxide)
- Reaction 1: 2NH₄Cl + CaCO₃ → CaCl₂ + 2NH₃ + H₂O + CO₂
- Reaction 2: 2NH₄Cl + Mg(OH)₂ → MgCl₂ + 2NH₃ + 2H₂O

A is incorrect as magnesium hydroxide is also a **base** and therefore causes **NH₃** release.

C & D are incorrect as sodium sulfate is not a base, it is a neutral salt, and does not react with ammonium salts.

10

The correct answer is **D** because:

- Nitrogen oxides are formed in car engines when nitrogen and oxygen taken from the air are reacted at the very high temperatures of the engine.
- Harmful nitrogen oxides are **returned to N₂ and O₂**, before being released into the atmosphere, by catalytic convertors:
 - **$2\text{NO} \rightarrow \text{N}_2 + \text{O}_2$ and $2\text{NO}_2 \rightarrow \text{N}_2 + 2\text{O}_2$**
- **Increasing the surface area** of the catalyst provides a larger reaction surface and therefore **more reactions** using the catalyst can occur ensuring **more** nitrogen oxides are converted into N₂ and O₂ – **decreasing the nitrogen dioxide concentration**.

A & C are incorrect as **increasing** the flow of exhaust gases will **decrease the time** available for the reaction to take place, so **less catalytic reactions** will take place and **increasing** the concentration of nitrogen dioxides. **Higher temperatures** will increase the energy available for the initial N₂ + O₂ reaction to take place resulting in a **higher concentration of nitrogen oxides** produced.

B is incorrect as **increasing** the flow of exhaust gases will **decrease the time** available for the reaction to take place, so **less catalytic reactions** will take place and **increasing** the concentration of nitrogen dioxides.

